MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

The wavelength of K_{α} X-ray produced by a X-ray tube is $0.76\tilde{A}\ldots$. The atomic number of the anode material of the tube is

(A) 38

(B) 41

(C) 63

(D) 53

Q.2

The position vector of a particle is given as $\overrightarrow{\mathrm{r}}=\left(\mathrm{t}^2-4t+6\right)\hat{\mathrm{i}}+\left(\mathrm{t}^2\right)\hat{\mathrm{j}}$. The time after which the velocity vector and acceleration vector becomes perpendicular to each other is equal to

(A) $1\mathrm{sec}$

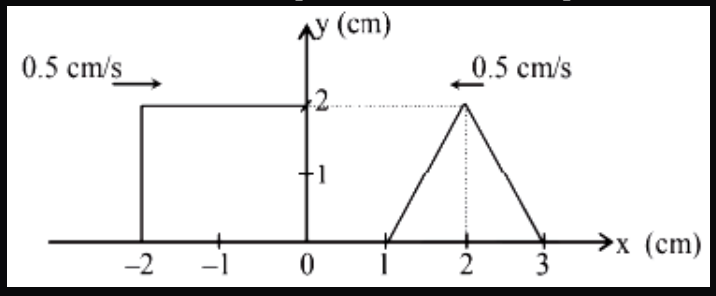
(B) $2\mathrm{sec}$

(C) $1.5\mathrm{sec}$

(D) not possible

Q.3

Figure shows a rectangular pulse and a triangular pulse approaching each other along x-axis. The pulse speed is $0.5\ \mathrm{cm/s}$. What is the resultant displacement of medium particles due to superposition of waves at $\mathrm{x}=0.5\ \mathrm{cm}$ and $\mathrm{t}=2\mathrm{sec}$.



(A) $3.5\ \mathrm{cm}$

(B) $2.5\ \mathrm{cm}$

(C) $4\ \mathrm{cm}$

(D) $3\ \mathrm{cm}$

Q.4

I is the image of a point object O formed by spherical mirror, then which of the following statement is incorrect:

(A) If O and I are on same side of the principal axis, then they have to be on opposite sides of the mirror.

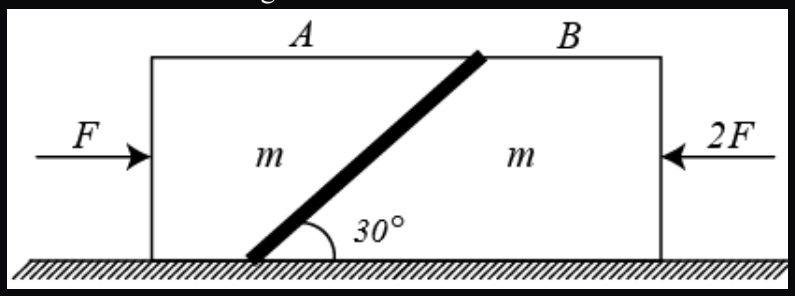
(B) If O and I are on opposite side of the principal axis, then they have to be on same side of the mirror.

(C) If O and I are on opposite side of the principal axis, then they can be on opposite side of the mirror as well.

(D) If O is on principal axis then I has to lie on principal axis only.

Q.5

Two blocks ' A ' and ' B ' each of mass ' m ' are placed on a smooth horizontal surface. Two horizontal force F and $2F$ are applied on both the blocks ' A ' and ' B ' respectively as shown in figure. The block A does not slide on block B. Then the normal reaction acting between the two blocks is :



(A) F

(B) $F/2$

(C) $\frac{F}{\sqrt{3}}$

(D) $3\ \mathrm{F}$

Q.6

If first excitation potential of a hydrogen like atom is V electron volt, then the ionization energy of this atom will be:

(A) V electron volt

(B) $\frac{3\ \mathrm{V}}{4}$ electron volt

(C) $\frac{4\ \mathrm{V}}{3}$ electron volt

(D) cannot be calculated by given information,

Q.7

A particle is thrown upwards from ground. It experiences a constant resistance force which can produce retardation of $2\ \mathrm{m/s^2}$. The ratio of time of ascent to the time of descent is :
$\left[\mathrm{g} = 10\ \mathrm{m/s^2}\right]$

(A) $1:1$

(B) $\sqrt{\frac{2}{3}}$

(C) $\frac{2}{3}$

(D) $\sqrt{\frac{3}{2}}$

Q.8

$\frac{E^2}{\mu_0}$ has the dimensions (E = electric field, μ_0 = permeability of free space)

(A) $\left[M^2L^3T^{-2}A^2\right]$

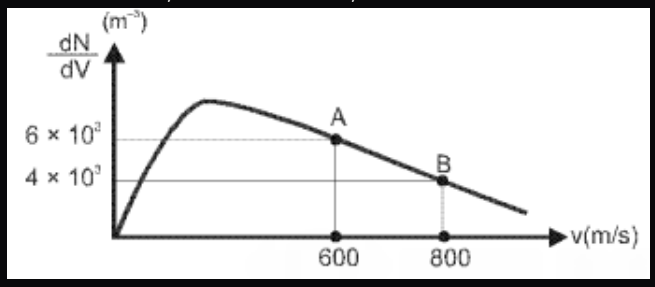
(B) $\left[\mathrm{MLT}^{-4}\right]$

(C) $\left[\mathrm{ML}^3\ \mathrm{T}^{-2}\right]$

(D) $\left[M^{-1}L^2TA^{-2}\right]$

Q.9

Maxwells speed distribution curve for an ideal gas is shown in the figure. If the number of the molecules in the velocity range from $600\ \mathrm{m/s}$ and $800\ \mathrm{m/s}$ is 10^{n}, find the value of n. Assume that the graph between the points A and B is linear.



(A) 3

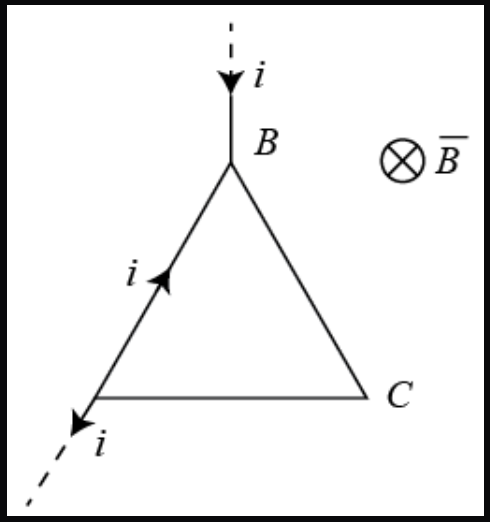
(B) 6

(C) 5

(D) 2

Q.10

Figure shows an equilateral triangle ABC of side ℓ carrying currents, placed in uniform magnetic field B. The magnitude of magnetic force on triangle is



(A) $il B$

(B) $2il B$

(C) $3i\ell B$

(D) zero

Q.11

There are four objects A, B, C and D. It is observed that A and B are in thermal equilibrium and C and D are also in thermal equilibrium. However, A and C are not in thermal equilibrium. We can conclude that:

(A) B and D are in thermal equilibrium

(B) B and D could be in thermal equilibrium

(C) B and D cannot be in thermal equilibrium

(D) The zeroth law of thermodynamics does not apply here because there are more than three objects

Q.12

If the short wavelength limit of the continuous spectrum coming out of a coolidge tube is $10\tilde{A}\ldots$, then the debroglie wavelength of the electrons reaching the target metal in the coolidge tube is approximately

(A) $0.3\tilde{A}\ldots$

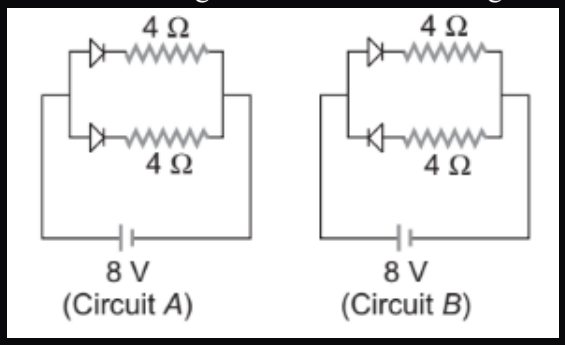
(B) 3 Ã…

(C) $30\tilde{A}\ldots$

(D) $10\tilde{A}\ldots$

Q.13

Currents flowing in each of the following circuits A and B respectively are



(A) 1 A, 2 A

(B) 2 A, 1 A

(C) 4 A, 2 A

(D) 2 A, 4 A

Q.14

On an X temperature scale, water freezes at $-125.0^{\circ}\mathrm{X}$ and boils at $375.0^{\circ}\mathrm{X}$. On a Y temperature scale, water freezes at $-70.0^{\circ}\mathrm{Y}$ and boils at $-30.0^{\circ}\mathrm{Y}$. The value of temperature on X-scale equal to the temperature of $50.0^{\circ}\mathrm{Y}$ on Y-scale is :

(A) $455.0^{\circ}\mathrm{X}$

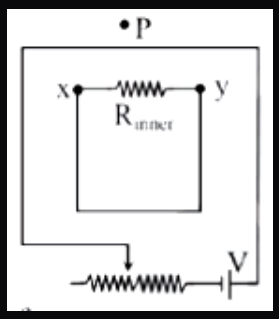
(B) $-125.0^{\circ}\mathrm{X}$

(C) $1375.0^{\circ}\mathrm{X}$

(D) $1500.0^{\circ}\mathrm{X}$

Q.15

For the figure shown, the variable resistance of the outer circuit R_{outer}, , is decreasing at a constant rate. While this is occurring in which direction is the magnetic field associated with the inner circuit at the point P in the plane of circuit & in which direction is induced current through the resistor R_{inner} in the smaller interior circuit. The two circuits lie on a flat table -



(A) Magnetic field into the page & current from x to y

(B) Magnetic field into the page & current from y to x

(C) Magnetic field out of the page & current from x to y

(D) Magnetic field out of the page & current from y to x

Q.16

A body of mass $32\ \mathrm{kg}$ is suspended by a spring balance from the roof of a vertically operating lift and going downward from rest. At the instants the lift has covered $20\ \mathrm{m}$ and $50\ \mathrm{m}$, the spring balance showed $30\ \mathrm{kg}$ & $36\ \mathrm{kg}$ respectively. The velocity of the lift is:

(A) decreasing at $20\ \mathrm{m}$ & increasing at $50\ \mathrm{m}$

(B) increasing at $20\ \mathrm{m}$ & decreasing at $50\ \mathrm{m}$

(C) continuously decreasing at a constant rate throught the journey

(D) continuously increasing at constant rate throughout the journey

Q.17

Binary stars rotate under mutual gravitational force at separation $2\left(\frac{G}{\omega^2}\right)^{\frac{1}{3}}$, where ω is the angular velocity of each of the star about centre of mass of the system. If difference between the mass of stars is 6 units. Find the ratio of mass of bigger star to smaller star.

(A) 7

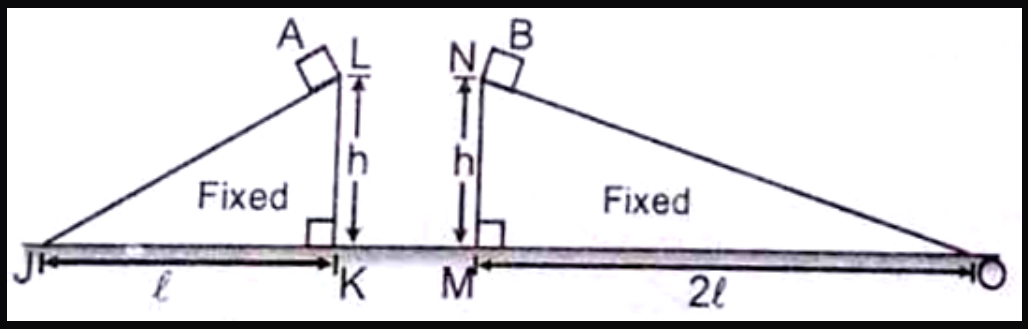
(B) 6

(C) 5

(D) 2

Q.18

Two identical blocks A and B are placed on two inclined planes as shown in diagram. Neglect air resistance and other friction



Read the following statements and choose the correct options.
Statements I : Kinetic energy of 'A' on sliding to J will be greater than the kinetic energy of B on falling to M.
Statements II : Acceleration of ' A ' will be greater than acceleration of ' B ' when both are released to slide on inclined plane
Statements III : Work done by external agent to move block slowly from position B to O is negative

(A) only statement I is true

(B) only statement II is true

(C) only I and III are true

(D) only II and III are true

Q.19

The electric field in a region is given by $\vec{E}=\frac{E_0 x}{l}\hat{i}$. The charge contained inside a cubical volume bounded by the surfaces $\mathrm{x}=0, \mathrm{x}=l, \mathrm{y}=0, \mathrm{y}=l, \mathrm{z}=0$ and $\mathrm{z}=l$ is -

(A) $2\varepsilon_0 \mathrm{E}_0 l^2$

(B) $\varepsilon_0 \mathrm{E}_0 l^2$

(C) $6\varepsilon_0 \mathrm{E}_0 l^2$

(D) zero

Q.20

A ray of light passes through an equilateral prism such that the angle of incidence is equal to the angle of emergence and each one is equal to $3/4$ th the angle of prism. The angle of deviation is

(A) 45°

(B) 39°

(C) 20°

(D) 30°

Q.21

There are three persons A, B \& C moving with constant velocity. Speed of A is $10\ \mathrm{m/sec}$ towards east, velocity of B relative to A is $6\ \mathrm{m/sec}$ at an angle of $\cos^{-1}\left(\frac{15}{24}\right)$ north of east. The velocity of C relative to B is $12\ \mathrm{m/sec}$ towards west. What will be the magnitude of velocity of C in $\mathrm{m/sec}$?

Q.22

The number of turns, cross-sectional area and length for four solenoids are given in the following table.

Solenoid	Total Turns	Area	Length
1	2 N	$2A$	ℓ
2	2 N	A	ℓ
3	3 N	$3A$	2ℓ
4	2 N	$2A$	$\ell/2$

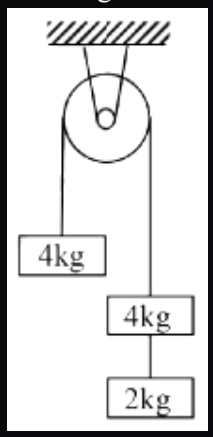
The solenoid with maximum self inductance is :

Q.23

The equation of motion of a particle of mass $1\ \mathrm{g}$ is $\frac{\mathrm{d}^2\mathrm{x}}{\mathrm{dt}^2} + \pi^2\mathrm{x} = 0$ where x is displacement (in m)from mean position. The frequency of oscillation is (in Hz) w, find $4\mathrm{w}$.

Q.24

The magnitude of acceleration of centre of mass of given system is $N/10\ \mathrm{m/s^2}$. Find N.

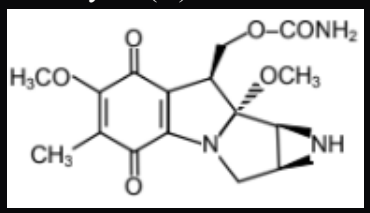


Q.25

The magnetic flux through metal ring varies with time t according to $\phi = 3\left(\mathrm{at}^3 - \mathrm{bt}^2\right)\ \mathrm{Wb}$. with $\mathrm{a} = 2\mathrm{sec}^{-3}$ and $\mathrm{b} = 6\mathrm{sec}^{-2}$. The resistance of the ring is 3Ω. Determine the maximum current induced in the ring during internal from $t = 0$ to $t = 2\mathrm{sec}$. (Mark absolute value as answer)

Q.26

Mitomycin (A)



How many number of sp^2 hybridized carbon atom in the structure of mitomycine (A)?

(A) 7

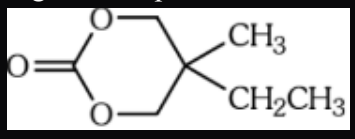
(B) 4

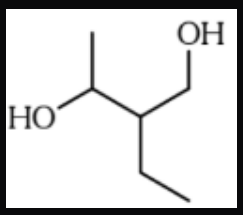
(C) 8

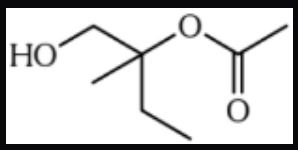
(D) 9

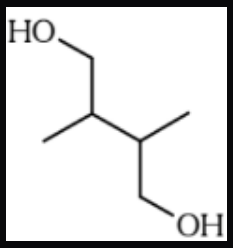
Q.27

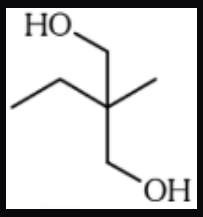
On being heated with a dilute solution of sulfuric acid in water, the compound shown was observed to liberate CO_2. What organic compound is formed during the reaction?



(A) 

(B) 

(C) 

(D) 

Q.28

Out of molar entropy (I), specific volume (II), heat capacity (III) and volume (IV), extensive properties are

(A) I, II

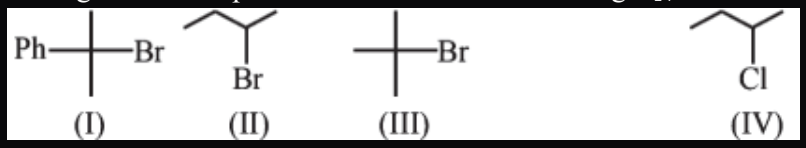
(B) I, II, IV

(C) II, III

(D) III, IV

Q.29

Arrange these compounds in the order of increasing S_N1 reaction rate -



(A) IV $<$ II $<$ III $<$ I

(B) I $<$ II $<$ III $<$ IV

(C) IV $<$ I $<$ III $<$ II

(D) IV $<$ I $<$ II $<$ III

Q.30

The reaction $[Fe(CN)_6]^{3-} \rightarrow [FeF_6]^{3-}$ takes place with :

(A) Decrease in magnetic moment

(B) Increase in magnetic moment

(C) Decrease in co-ordination number

(D) Increase in co-ordination number

Q.31

Sodium fusion extract of aniline when heated with ferrous sulphate solution and then acidified with concentrated H_2SO_4 from which of the following complexes?

(A) $[Fe(CN)_6]^{4-}$

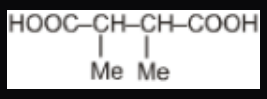
(B) $Fe_4\,[Fe(CN)_6]_3 \cdot xH_2O$

(C) $[Fe(SCN)]^{2+}$

(D) $[Fe(CN)_5NOS]^{4-}$

Q.32

The most stable form of meso



in strong basic medium is :

(A) Anti form

(B) Gauche form

(C) Fully eclipsed from

(D) Partially eclipsed form

Q.33

Which of the following is correct?

(A) $D(+)$-glucose and $D(+)-$ mannose are enantiomers.

(B) $\mathrm{D}(+)-$ glucose and $\mathrm{D}(-)-$ fructose give same osazone.

(C) $\mathrm{D}(-)$-fructose give single product on reduction with $\mathrm{Na}-\mathrm{Hg}$.

(D) $\mathrm{D}(+)$-glucose and $\mathrm{D}(-)$-fructose are epimers.

Q.34

Which of the following statements are correct for classification of elements?

I. The properties of elements are periodic functions of their atomic numbers.

II. Non-metallic elements are less in number than the metallic elements.

III. The first ionisation energies of elements along a period do not vary in a regular manner.

IV. The ground state electronic configuration of $\mathrm{Pd}(Z=46)$ is $[\mathrm{Kr}]4d^{8}5s^{2}$.

(A) I, II, III, IV

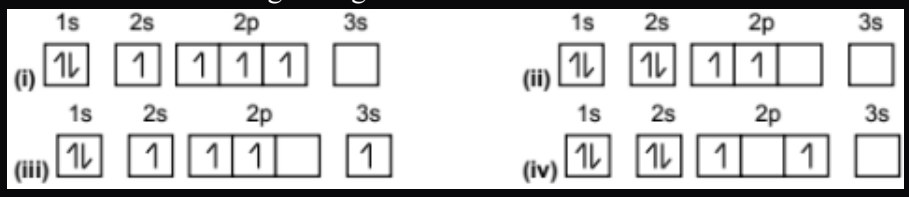
(B) I, II, III only

(C) II, III, IV only

(D) I, II, IV only

Q.35

Consider the following configurations:



Which of the above configuration may represent the excited state of C-atom.

(A) (i), (iii) & (iv)

(B) (i) only

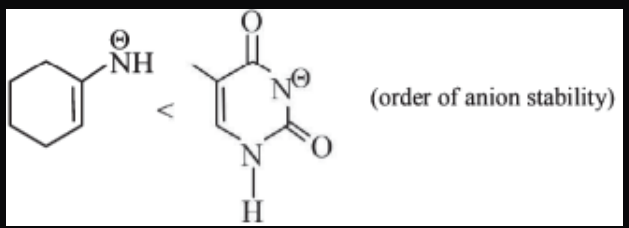
(C) (i) & (iii)

(D) (iii) only

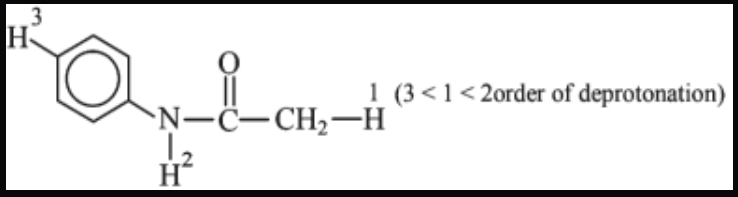
Q.36

Identify the incorrect option according to the property given in the parenthesis-

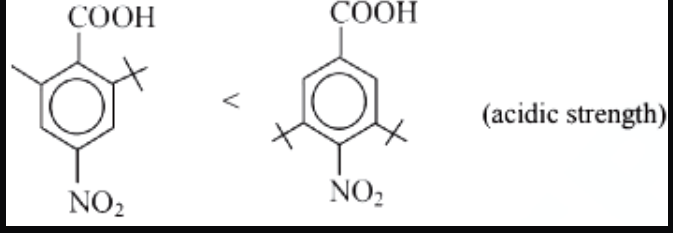
(A)



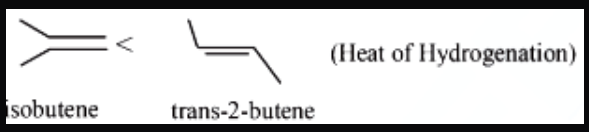
(B)



(C)



(D)



Q.37

Which of the following sets is in the correct order regarding the property mentioned against them?

	Sets	Property
I.	$NCCH_2COOH > FCH_2COOH > H_3CCH_2COOH$	Acidity
II.	$CH_3CH_2CHO > PhCOCH_3 > PhCHO$	Reactivity
III.	$H_3COCH_2CH_3 < H_3CCH_2CHO < H_3CCH_2CH_2OH$	Boiling points

(A) I, II only

(B) I, III only

(C) II, III only

(D) I, II, III

Q.38

Match the following?

property	metal
(a) Element with highest second ionization enthalpy (Δ_iH_2)	(i) Co
(b) Element with highest third ionization enthalpy (Δ_iH_3)	(ii) Cr
(c) M in $[M(CO)_6]$	(iii) Cu
(d) Element with highest heat of atomization (Δ_aH)	(iv) Zn
	(v) Ni

(A) $(a \rightarrow \text{iii}),(b \rightarrow i),(c \rightarrow v),(d \rightarrow iv)$

(B) $(a \rightarrow iii), (b \rightarrow iv), (c \rightarrow ii), (d \rightarrow v)$

(C) $(a \rightarrow iv), (b \rightarrow i), (c \rightarrow ii), (d \rightarrow iii)$

(D) $(a \rightarrow v), (b \rightarrow iv), (c \rightarrow i), (d \rightarrow iii)$

Q.39

Two radio isotopes P and Q of atomic mass 10 and 20 respectively are mixed in equal amount by mass. After 20 days, their weight ratio $\frac{W_P}{W_Q}$ is found to be $1:4$. Isotope Q has a half-life of 10 days. The half-life of isotope P is:

(A) Zero

(B) 5 days

(C) 20 days

(D) Infinite

Q.40

The e.m.f. of a cell corresponding to the reaction
$\mathrm{Zn}+2\mathrm{H}^{+}(\mathrm{aq}) \longrightarrow \mathrm{Zn}^{2+}(0.1\mathrm{M})+\mathrm{H}_2(\mathrm{~g})(1 \mathrm{~atm})$
is 0.26 volt at $25^{\circ} \mathrm{C}$. Calculate the pH of the solution at the hydrogen electrode.
$\mathrm{E}^{\circ}_{\mathrm{Zn}^{2+}/\mathrm{Zn}}=-0.76$ volt and $\mathrm{E}^{\circ}_{\mathrm{H}^{+}/\mathrm{H}_2}=0$

(A) 8

(B) 9

(C) 1

(D) 2

Q.41

In $2\mathrm{HI}(\mathrm{g}) \rightleftharpoons \mathrm{H}_2(\mathrm{~g})+\mathrm{I}_2(\mathrm{~g}), 2 \mathrm{~mole} \mathrm{HI}, 4$ mole H_2 and 4 mole of I_2 are found at equilibrium. If some hydrogen is now introduced into the container such that at equilibrium concentration of HI and I_2 become equal. Find the Number of mole of H_2 added.

(A) 10 mole

(B) 20 mole

(C) 1 mole

(D) None of these

Q.42

Cr^{2+} and Mn^{3+} do possess d^4 electronic configuration. So,

(A) Mn^{3+} is oxidizing agent while Cr^{2+} is reducing agent

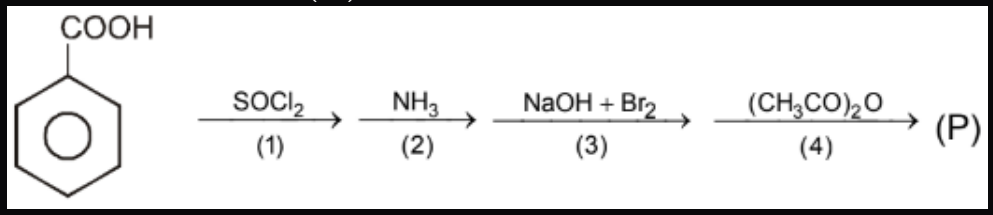
(B) Both are reducing agents

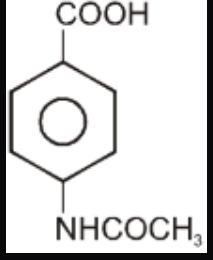
(C) Mn^{3+} is reducing agent while Cr^{2+} is oxidizing agent

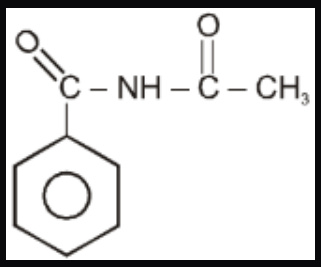
(D) Both are oxidizing agents

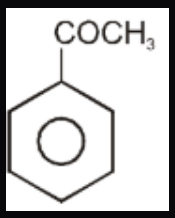
Q.43

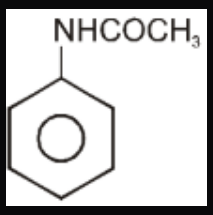
The structure of product (P) is :



(A) 

(B) 

(C) 

(D) 

Q.44

Given $E^{\circ}_{Cr^{3+}/Cr} = -0.72\ V, E^{\circ}_{Fe^{2+}/Re} = -0.42\ V$
The potential for the cell $Cr\left|Cr^{3+}(0.1M)\right|\left|Fe^{2+}(0.01M)\right|Fe$ is

(A) $0.26\ V$

(B) $0.399\ V$

(C) $-0.339\ V$

(D) $-0.26\ V$

Q.45

The IUPAC name of an unsymmetrical ether with the molecular formula $= C_4H_{10}O$

(A) ethoxy propane

(B) methoxy ethane

(C) ethoxy ethane

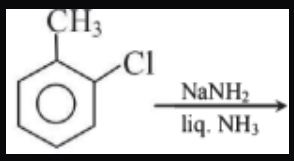
(D) methoxy propane

Q.46

A solution containing 0.1 g of a non-volatile organic substance P (molecular mass 100) in 100 g of benzene raises the

boiling point of benzene by $0.2^{\circ}\mathrm{C}$, while a solution containing $0.1\ \mathrm{g}$ of another non-volatile substance Q in the same amount of benzene raises the boiling point of benzene by $0.4^{\circ}\mathrm{C}$. What is the ratio of molecular masses of P and Q in term of P/Q ?

Q.47



In above reaction how many isomeric product will be formed? (Benzyne mechanism)

Q.48

How many isomeric dienes with a six membered ring are possible of the compound with the molecular formula $\mathrm{C_7H_{10}}$? (Hint : Two consecutive double bonds are not possible in six membered ring).

Q.49

Calculate total number of secondary amine (only structural isomers) with molecular formula $\mathrm{C_5H_{13}\ N}$?

Q.50

Speed of electron in 4th orbit of a hydrogen like ion is $2.725 \times 10^6\ \mathrm{m/sec}$. Give atomic number of ion.

Q.51

Find the area bounded by the curves, $x^2+y^2=25, 4y=\left|4-x^2\right|$ and $x=0$ in the first quadrant.

(A) $1+\sin^{-1}\left(\frac{4}{5}\right)$

(B) $2+25\sin^{-1}\left(\frac{4}{5}\right)$

(C) $2+\frac{25}{2}\sin^{-1}\left(\frac{4}{5}\right)$

(D) None of these

Q.52

$f: R \rightarrow R$ is defined as $f(x)=\begin{cases} x^2+2mx-1 & \text{for } x \leq 0 \\ mx-1 & \text{for } x>0 \end{cases}$.

If $f(x)$ is one-one then m must lies in the interval

(A) $(-\infty, 0)$

(B) $(-\infty, 0]$

(C) $(0, \infty)$

(D) $[0, \infty)$

Q.53

Let A & B are two independent events such that $\mathrm{P(A)+P(B)}=\frac{3}{4}$ & $\mathrm{P}(\overline{\mathrm{A}}/\mathrm{B})=\frac{2}{5}$, then $\mathrm{P(A \cap B)}$ is-

(A) $\frac{9}{20}$

(B) $\frac{9}{100}$

(C) $\frac{5}{17}$

(D) $\frac{6}{20}$

Q.54

If a, b, c are non zero real numbers, then minimum value of the expression $\left(\frac{\left(a^4+a^2+1\right)\left(b^4+7b^2+1\right)\left(c^4+11c^2+1\right)}{a^2b^2c^2}\right)$, , is -

(A) 315

(B) 351

(C) 415

(D) 451

Q.55

If p_1, p_2 are the roots of the quadratic equation $ax^2 + bx + c = 0$ and q_1, q_2 are the roots of the quadratic equation $cx^2 + bx + a = 0 (a, b, c \in R)$ such that p_1, q_1, p_2, q_2 are in A.P. of distinct terms, then $\frac{a}{c}$ equals

(A) -1

(B) 1

(C) $\frac{1}{2}$

(D) 2

Q.56

$\text{Lim}_{x\to 1}\ \frac{\sin^2\left(x^3+x^2+x-3\right)}{1-\cos\left(x^2-4x+3\right)}$ has the value equal to

(A) 18

(B) $9/2$

(C) 9

(D) None

Q.57

Let X be the solution set of the equation $A^x = I$, where $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ and I is the unit matrix and $X \subset N$ then the minimum value of $\sum_x \left(\cos^x \theta + \sin^x \theta\right), \theta \in R$ is :

(A) 1

(B) 2

(C) 4

(D) -2

Q.58

The equation of the image of the pair of rays $y = |x|$ by the line $y = 1$, is

(A) $y = -|x| + 2$

(B) $y = |x| - 2$

(C) $y = |x| + 1$

(D) $y = |x| - 1$

Q.59

Let A & B are two non singular matrices of order 3 such that $A + B = I \& A^{-1} + B^{-1} = 2I$, then $|\operatorname{adj}(4AB)|$, is (where $\operatorname{adj}(A)$ is adjoint of matrix A)-

(A) 4

(B) 16

(C) 64

(D) 128

Q.60

Solution of differential equation $x^2 \frac{dy}{dx} + y^2 e^{\frac{x(y-x)}{y}} = 2y(x-y)$ be given by
Where C is constant of integration

(A) $x(x+y) = y \log\left(\mathrm{Ce}^x - 1\right)$

(B) $x(x-y) = x \log\left(\mathrm{Ce}^x - 1\right)$

(C) $x(x+y) = x \log\left(\mathrm{Ce}^x + 1\right)$

(D) $x(x-y) = y \log\left(\mathrm{Ce}^x - 1\right)$

Q.61

Let $f(x) = x^2 - x + k - 2, k \in R$, then the complete set of values of k for which $\mathrm{y} = |\mathrm{f}(|\mathrm{x}|)|$ is non-derivable at 5 distinict points is

(A) $(1, 4)$

(B) $\left(0, \frac{9}{4}\right)$

(C) $(-\infty, 2)$

(D) $\left(2, \frac{9}{4}\right)$

Q.62

If α, β, γ are the roots of $x^3 + x + 3 = 0$, then find the value of $\left|\prod\left(\frac{1-\alpha}{1+\alpha}\right)\right|$. (where Π means continued product)

(A) 5

(B) 4

(C) 6

(D) -4

Q.63

Let $\mathrm{a}_1, \mathrm{a}_2, \ldots \ldots \mathrm{a}_{101}$ is a group of real numbers such that $a_i > a_{i+1}$ for all values of i and mean square deviation of the group is minimum about the number ' a_{51} ', then mode of the group will be -

(A) $2\mathrm{a}_{51}$

(B) a_{51}

(C) a_{50}

(D) a_{52}

Q.64

Let f be a derivable function satisfying the equation $\int_0^{\mathrm{x}} f(\mathrm{t})\mathrm{dt} + \int_0^{\mathrm{x}} \mathrm{t} \cdot f(\mathrm{x}-\mathrm{t})\mathrm{dt} = \mathrm{e}^{-\mathrm{x}} - 1$.
$\int_0^1 f(x)dx$ is equal to-

(A) $\mathrm{e} - 1$

(B) -1

(C) 1

(D) $-\mathrm{e}^{-1}$

Q.65

Let $A \equiv (\lambda + 2, 1 - 2\lambda, \lambda + 2)$ and $B \equiv (2k + 1, k, k + 1)$ and $\lambda, k \in R$. Then minimum distance between A and B is -

(A) 0

(B) $\frac{1}{\sqrt{35}}$

(C) $\frac{\sqrt{3}}{\sqrt{35}}$

(D) $\frac{3}{\sqrt{35}}$

Q.66

Let $\cos^{-1}(x) + \cos^{-1}(2x) + \cos^{-1}(3x) = \pi$. If x satisfies the cubic $ax^3 + bx^2 + cx - 1 = 0$, then $(a + b + c)$ has the value equal to

(A) 24

(B) 25

(C) 26

(D) 27

Q.67

Let $ABCD$ is a parallelogram where $\overrightarrow{\mathrm{AB}} = \overrightarrow{\mathrm{a}}, \overrightarrow{\mathrm{AD}} = \overrightarrow{\mathrm{b}}, |\overrightarrow{\mathrm{a}}| = |\overrightarrow{\mathrm{b}}| = 2$ and $|\vec{a} \times \vec{b}| + \vec{a} \cdot \vec{b} = \sqrt{2}|\vec{a}||\vec{b}|(\vec{a} \cdot \vec{b} > 0)$, then area of this parallelogram, is (in square units)-

(A) $2\sqrt{2}$

(B) 2

(C) $\sqrt{2}$

(D) $8\sqrt{2}$

Q.68

$A(1,0)$ and $B(0,1)$ and two fixed points on the circle $x^2+y^2=1$. C is a variable point on this circle. As C moves, the locus of the orthocentre of the triangle ABC is

(A) $x^2+y^2-2x-2y+1=0$

(B) $x^2+y^2-x-y=0$

(C) $x^2+y^2=4$

(D) $x^2+y^2+2x-2y+1=0$

Q.69

The minimum value of
$f(x)=|x-1|+|2x-1|+|3x-1|+\ldots\ldots+|119x-1|$
occurs at x. Then x is equal to-

(A) $\frac{1}{84}$

(B) $\frac{1}{51}$

(C) $\frac{1}{80}$

(D) $\frac{1}{94}$

Q.70

Solution of differential equation
$\left(1+\mathrm{e}^{2y}\right)\mathrm{e}^{\tan^{-1}\mathrm{x}}\mathrm{dx}-\left(1+\mathrm{x}^2\right)\left(\mathrm{e}^{\mathrm{y}}+\left(\mathrm{e}^{\mathrm{y}}-1\right)^2\right)\mathrm{dy}=0$
is-
(where C is constant of integration)

(A) $\ln(\mathrm{y})=\tan\left(\mathrm{y}-\mathrm{e}^{\tan^{-1}\mathrm{x}}+\mathrm{C}\right)$

(B) $y=\ln\left(\tan\left(y-e^{\tan^{-1}x}+C\right)\right)$

(C) $\ell\ln(\mathrm{y})=\tan\left(\mathrm{e}^{\tan^{-1}\mathrm{x}}-\mathrm{y}+\mathrm{C}\right)$

(D) $y=\ln\left(\tan\left(e^{\tan^{-1}x}-y+C\right)\right)$

Q.71

If $\lim_{x\to0^+}\frac{\{x\}^{\{x\}}\cos\left(\{x\}^{\{\sin x\}+1}\right)}{\cos(\tan\{x\}-x)\cos(\{x+1\}-x)}=k$, then k is equal to (where $\{.\}$ denotes fractional part function)

Q.72

The last two digits of $2015!+3^{2015}$ are XY, find X+Y.

Q.73

Let $L_1 : \frac{x-1}{1} = \frac{y-1}{2} = \frac{z-1}{k}$ and $L_2 : \frac{x-3}{1} = \frac{y-2}{-1} = \frac{z-3}{-1}$ be two intersecting lines and L_3 be a straight line which is perpendicular to L_1 and L_2 and passing through their point of intersection. If length of perpendicular from $(3, 1, 2)$ to the line L_3 is λ then $[\lambda]$ equals [Note: $[\mathrm{y}]$ denotes the greatest integer function less than or equal to y.]

Q.74

If complex numbers z_1 and z_2 both satisfy $\mathrm{z} + \overline{\mathrm{z}} = 2|\mathrm{z} - 1|$ and $\arg(\mathrm{z}_1 - \mathrm{z}_2) = \frac{\pi}{3}$, then value of $\mathrm{Im}(z_1 + z_2)$ is, where $\mathrm{Im}(z)$ denotes imaginary part of z is $K/\sqrt{3}$, find K.

Q.75

If $I_1 = \int_0^{\pi} \frac{\sin 884x \sin 1122x}{\sin x} dx$ & $I_2 = \int_0^1 \frac{x^{238}\left(x^{1768}-1\right)}{x^2-1} dx$, then $\frac{I_1}{I_2}$ is equal to

Answers & Solutions

Q.1 Answer:

41

Solution:

$n = 2$ to $n = 1$
$z_{\text{eff}} = (z-1)$
$\frac{1}{\lambda} = R\left\{\frac{1}{1^2} - \frac{1}{2^2}\right\}(z-1)^2$
$\frac{10^{10}}{0.76} = 1.09 \times 10^7 \times \frac{3}{4}(z-1)^2$
$(z-1) = 40$
$z = 41$

Q.2 Answer:

1sec

Solution:

$\vec{r} = \left(t^2 - 4t + 6\right)\hat{i} + t^2\hat{j}; \quad \vec{v} = \frac{d\vec{r}}{dt} = (2t-4)\hat{i} + 2t\hat{j}, \quad \vec{a} = \frac{d\vec{v}}{dt} = 2\hat{i} + 2\hat{j}$

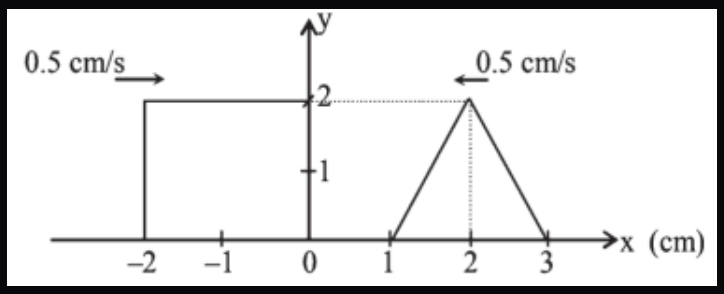
if a and $\bar{v}$ are perpendicular

$\vec{a} \cdot \vec{v} = 0$
$(2\hat{i} + 2\hat{j}) \cdot ((2t-4)\hat{i} + 2t\hat{j}) = 0$
$8\mathrm{t} - 8 = 0$
$t = 1$ sec.

Q.3 Answer:

3 cm

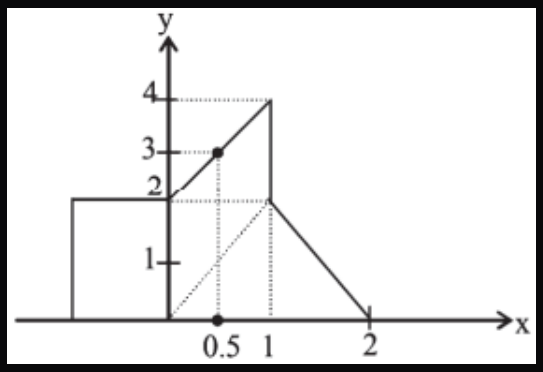
Solution:



$x_1 = 0.5 \times t = 0.5 \times 2 = 1 \text{ cm}$

$x_2 = 0.5 \times t = 0.5 \times 2 = 1 \text{ cm}$

$\text{at } t = 2\text{sec}$



at $x = 0.5 \text{ cm}, y = 3 \text{ cm}$

Q.4 Answer:

If O and I are on opposite side of the principal axis, then they can be on opposite side of the mirror as well.

Solution:

$\frac{I}{O} = -\frac{V}{u}$

If O and I are on same sides of PA. $\frac{I}{O}$ will be positive which implies v and u will be of opposite signs.

Similarly if O and I are on opposite sides, $\frac{I}{O}$ will be -ve which implies v and u will have same sign.

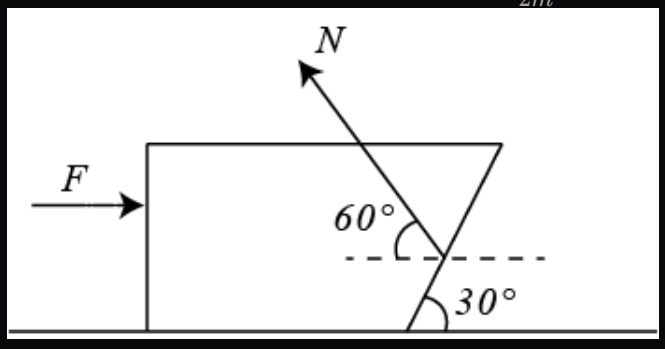
If O is on $PA, I = \left(-\frac{V}{u}\right)(O) \Rightarrow I$ will also be on. P.A.

Q.5 Answer:

3 F

Solution:

Acceleration of two mass system is a $= \frac{F}{2m}$ leftward FBD of block A



$N \cos 60^\circ - F = ma = \frac{mF}{2m}$ **solving,** $N = 3F$

Q.6 Answer:

$\frac{4V}{3}$ electron volt

Solution:

$$\text{First excitation energy } = \text{RhC}\left(\frac{1}{1^2}-\frac{1}{2^2}\right) = \text{RhC}\,\frac{3}{4}$$

$$\therefore \quad \frac{3}{4}\text{RhC} = \text{V e.v.} \quad \Rightarrow \quad \therefore \quad RhC = \frac{4V}{3}\text{ e.v.}$$

Q.7 Answer:

$\sqrt{\frac{2}{3}}$

Solution:

Let a be the retardation produced by resistive force, t_a and t_d be the time of ascent and descent respectively.
If the particle rises upto a height h then $h = \frac{1}{2}(g+a){t_a}^2$ and $h = \frac{1}{2}(g-a){t_d}^2$

$$\therefore \quad \frac{t_a}{t_d} = \sqrt{\frac{g-a}{g+a}} = \sqrt{\frac{10-2}{10+2}} = \sqrt{\frac{2}{3}} \quad \text{Ans. } \sqrt{\frac{2}{3}}$$

Q.8 Answer:

$\left[\text{MLT}^{-4}\right]$

Solution:

$$\left[\frac{E^2}{\mu_0}\right] = \left[\frac{\varepsilon_0 E^2}{\varepsilon_0\mu_0}\right] = \left[\frac{\text{energy/volume}}{(1/\text{ speed of light })^2}\right] = \left[\frac{\text{energy(speed)}^2}{\text{volume}}\right] = \left[\frac{\text{ML}^2T^{-2}L^2T^{-2}}{L^2}\right] = [MLT^{-4}$$

Q.9 Answer:

6

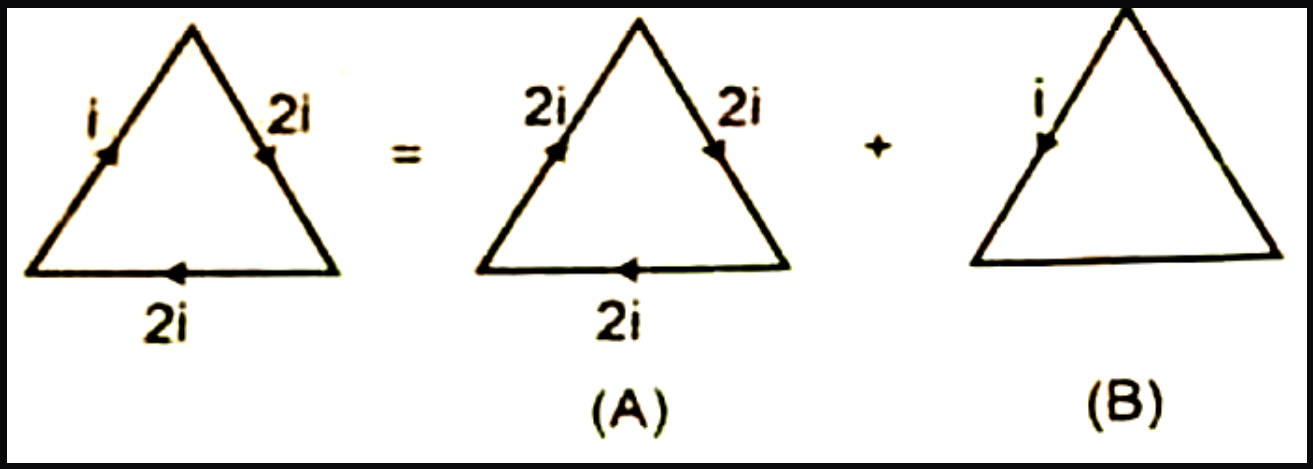
Solution:

No. of molecules = area under AB
$= \frac{(6+4)10^3}{2} \times (800-600) = 10^6$

Q.10 Answer:

$i\ell B$

Solution:



force in figure (A) is zero, and force in figure $(B) = i\ell B$.

Q.11 Answer:

B and D cannot be in thermal equilibrium

Solution:

Thermal equilibrium means same temperature.

Q.12 Answer:

$0.3\tilde{A}\ldots$

Solution:

We have : K.E. $= \frac{P^2}{2m_e} = \frac{hc}{\lambda_{\min}} \quad \Rightarrow \quad P = \sqrt{\frac{2hcm_e}{\lambda_{\min}}}$

Also, $\lambda_{\text{de broglie}} = \frac{h}{p} = \sqrt{\frac{h\lambda_{\min}}{2m_e c}}$

for $\lambda_{\min} = 10\tilde{A}\ldots : \lambda_{\text{de broglie}} \cong 0.3\tilde{A}\ldots$

Q.13 Answer:

4 A, 2 A

Solution:

In circuit A**, both** $(p-n)$ **junction diode act as forward biasing. Hence, current flows in circuit** A**.**

Total resistance R **is given by**

$\frac{1}{R} = \frac{1}{4} + \frac{1}{4}$

or $\quad \frac{1}{R} = \frac{2}{4}$

or $\quad R = 2\Omega$

According to Ohmâ€™s law

$$V = I_A R$$

or $\quad 8 = I_A \times 2$

or $\quad I_A = 4\ \text{A}$

Q.14 Answer:

1375.0°X

Solution:

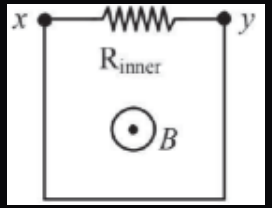
$\frac{X-(-125)}{500} = \frac{Y-(-70)}{40}$

For $Y = 50 \quad X = 1375.0^\circ X$

Q.15 Answer:

Magnetic field out of the page & current from x to y

Solution:



B **is increasing**

$\therefore \phi$ **is increasing**

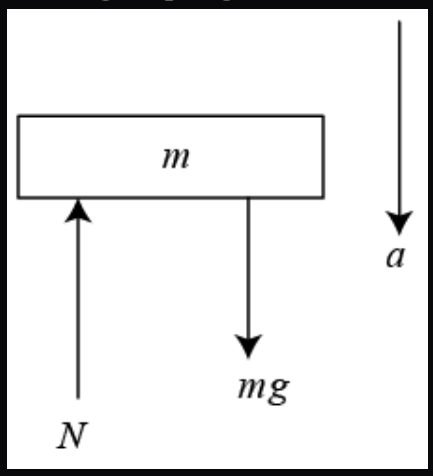
induced current is clockwise i is from x to y.
At P, B is outward

Q.16 Answer:

increasing at $20\ \mathrm{m}$ & decreasing at $50\ \mathrm{m}$

Solution:

$N = m(g - a), N < mg$ **if** $a(\downarrow)$
and $N > mg$ **if a** $(\uparrow)$
Reading of spring balance is less than m **if** $a(\downarrow)$ **and reading of spring balance is**



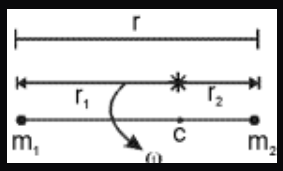
greater than m **if** $a(\uparrow)$

Q.17 Answer:

7

Solution:

Given that $\mathrm{m_1 - m_2 = 6}$ **unit ...(1)**



equation of motion
$\frac{Gm_1m_2}{r^2} = m_1\omega^2 r_1 = m_1\omega^2 \frac{m_2 r}{m_1+m_2}$
$\therefore \mathrm{m_1 + m_2} = \frac{\omega^2 r^3}{G} = 8\ \mathrm{unit.}$
by (1) and (2)
$\frac{m_1}{m_2} = 7$

Q.18 Answer:

only II and III are true

Solution:

Statement I : Work done by gravity is same for motion from A **to** J **and** B **to** M **for equal mass. So K.E. will be equal.**
Statement II : Acceleration $= g\sin\theta$

$\sin\theta_A > \sin\theta_B$

$\frac{h}{\ell} > \frac{h}{2\ell}$

Statement III :

$W_g + W_{\text{ext}} = 0$ (Because moved slowly)

$W_{\text{ext}} = -W^g$

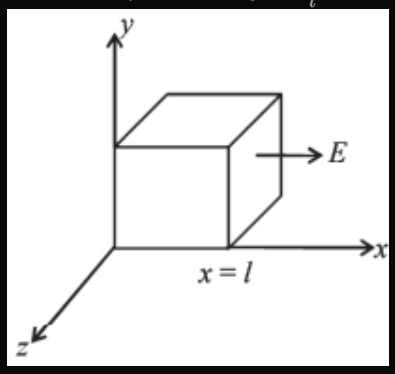
from B to O: W_g^{ext} is positive so $W_{\text{ext}} < 0$

Q.19 Answer:

$\varepsilon_0 \mathrm{E}_0 l^2$

Solution:

At $x = l, E = E_0 \times \frac{l}{l} = E_0$

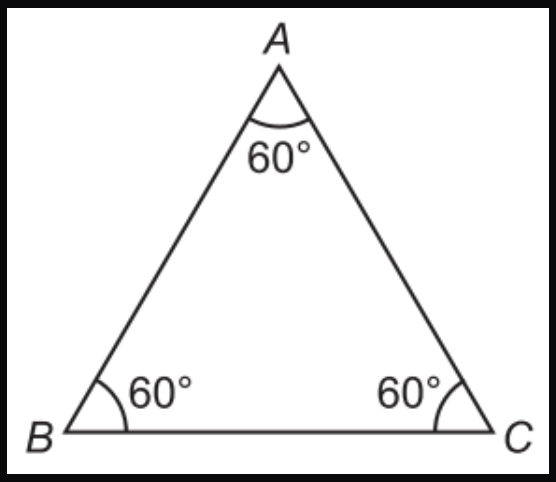


$\phi_{\text{total}} = \frac{q_{\text{in}}}{\varepsilon_0}; \quad \mathrm{E}_0 \cdot l^2 = \frac{q_{\text{in}}}{\varepsilon_0} \quad \therefore \mathrm{q}_{\text{in}} = \varepsilon_0 \mathrm{E}_0 l^2$

Q.20 Answer:

30°

Solution:



Relation among the $(i), (e), (A)$ and (δ)

$i + e = A + \delta$

$\frac{3}{4} \times 60^\circ + \frac{3}{4} \times 60^\circ = 60^\circ + \delta$

or $45^\circ + 45^\circ = 60^\circ + \delta$

or $\delta = 30^\circ$

where, i = angle of incidence

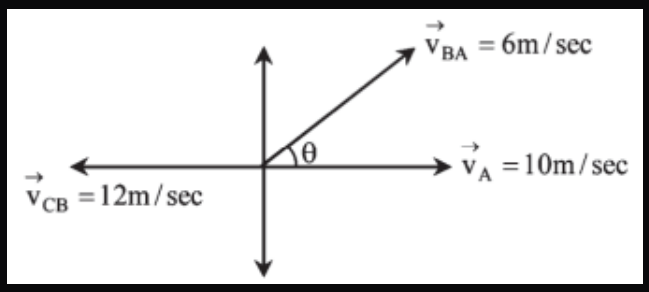
$e =$ angle of emergence

$A =$ angle of prism
$\delta =$ angle of deviation

Q.21 Answer:

—

Solution:



$$\cos\theta = \frac{15}{24}$$

$$\overrightarrow{v}_{BA} = \overrightarrow{v}_B - \overrightarrow{v}_A \Rightarrow \overrightarrow{v}_B = \overrightarrow{v}_{BA} + \overrightarrow{v}_A \ldots$$

$$\& \ \overrightarrow{v}_{CB} = \overrightarrow{v}_C - \overrightarrow{v}_B \Rightarrow \overrightarrow{v}_C = \overrightarrow{v}_{CB} + \overrightarrow{v}_B$$

From (i) & (ii)

$$\overrightarrow{v}_C = \overrightarrow{v}_{CB} + \overrightarrow{v}_A + \overrightarrow{v}_{BA}$$

Here $\vec{v}_{CB} + \vec{v}_A = 2 \text{ m/sec}$ **towards west**

Finally, $|\vec{v}_C| = \sqrt{(2)^2 + (6)^2 + 2(2)(6)\left(\frac{-15}{24}\right)}$

$= 5 \text{ m/sec}$

Q.22 Answer:

—

Solution:

$$L = \mu_0 n^2 \pi r^2 \ell$$

$$A = \pi r^2$$

$$n = \frac{N}{\ell} \quad \Rightarrow \quad L = \mu_0 \frac{N^2}{\ell} A$$

By putting the given values, it can be seen that it is maximum for solenoid no.4.

Q.23 Answer:

—

Solution:

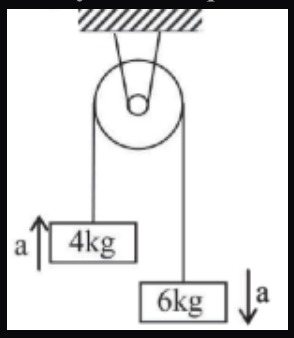
$$\omega^2 = \pi^2 \Rightarrow \omega = \pi \Rightarrow \text{f} = \frac{\omega}{2\pi} = \frac{\pi}{2\pi} = \frac{1}{2} \text{ Hz}$$

Q.24 Answer:

—

Solution:

The system is equivalent to



$$a = \frac{6-4}{6+4} \times g = 2\ m/s^2$$

$$\vec{a}_{com} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1+m_2} = \frac{(4)(2)+(6)(-2)}{4+6}$$

$$= \frac{8-12}{10}$$

$$\vec{a}_{com} = -\frac{4}{10} = -\frac{2}{5}\ m/s^2$$

$$\therefore \frac{2}{5} = \frac{N}{5}$$

$$N = 4$$

Q.25 Answer:

—

Solution:

$$e = -\frac{d\phi}{dt}$$

$$i = -\frac{1}{R}\frac{d\phi}{dt}$$

$$|i| = \frac{1}{3}\frac{d}{dt}\left(3at^3 - 3bt^2\right) = \frac{1}{3}\left[9at^2 - 6bt\right]$$

$$|i| = 3at^2 - 2bt = 6t^2 - 12t$$

For maxima of i

$$\frac{di}{dt} = 0$$

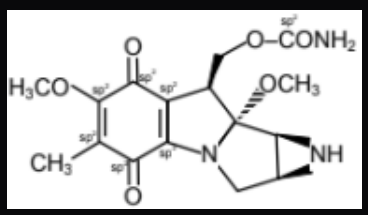
$$12t - 12 = 0 \Rightarrow t = 1sec$$

$$i_{max} = 6 \times 1^2 - 12 \times 1 = 6\ Amp.$$

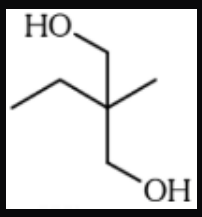
Q.26 Answer:

7

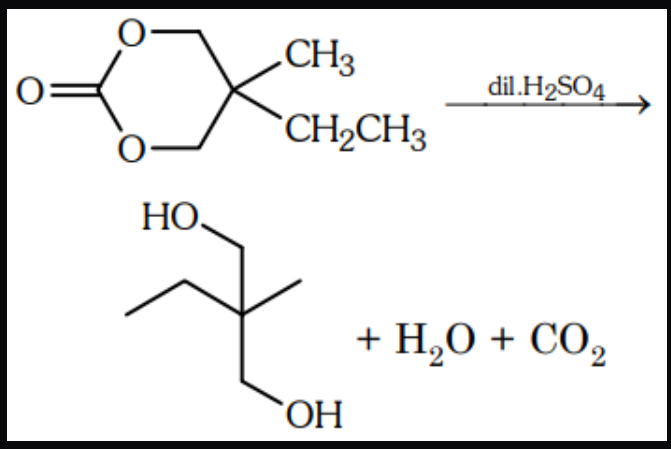
Solution:



Q.27 Answer:



Solution:



Q.28 Answer:

III, IV

Solution:

Heat capacity and volume are dependent on mass of the sample. So, they are extensive properties.

Q.29 Answer:

IV < II < III < I

Solution:

$S_N{}^C$ **reaction proceeds with carbocation so reactivity order is I > III > II > IV.**

Q.30 Answer:

Increase in magnetic moment

Solution:

CN^- **is strong field ligand due to which** Fe^{+3} **configuration will be** $t_{2\,g}^5$ **hence 1 unpaired electron whereas in** $[FeF_6]^{3-}$ **,** F^- **is weak field ligand hence due to weak field splitting, configuration will** $t_{2\,g}^3 e_g^2$ **i.e. 5 unpaired electrons.**

Q.31 Answer:

$Fe_4\,[Fe(CN)_6]_3 \cdot xH_2O$

Solution:

The carbon and nitrogen present in the organic compound of fusion with sodium metal give sodium cyanide $(NaCN)$ **soluble in water. This is converted into sodium ferrocyanide by the addition of sufficient quantity of ferrous sulphate. Ferric ions generated during the process react with ferrocyanide to form the prussian blue precipitate of ferric ferrocyanide. The prussian blue colour is due to formation of ferric ferrocyanide,**

$Fe_4\,[Fe(CN)_6]_4 \cdot X_2O.$

Reaction

(i) $Fe^{2+} + 6\,\overset{\ominus}{C}\,N \longrightarrow \left[Fe^{2+}(\overset{-6}{CN})_6\right]^{4-}$

Hexa cyanoferrate (II)

(ii) $Fe^{2+} \xrightarrow[H_2SO_4]{Conc.} Fe^{3+} + e^-$

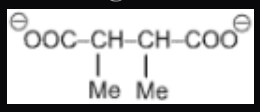
(iii) $\left[Fe^{2+}(\overset{6}{CN})_6\right]^{4-} + 4Fe^{3+} \longrightarrow \left[Fe^{2+}(\overset{6}{CN})_6\right]^{4-} \cdot H_2O$

Q.32 Answer:

Anti form

Solution:

In strong basic medium



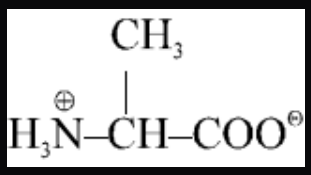
will be stable in its anti form to minimise e-repulsion

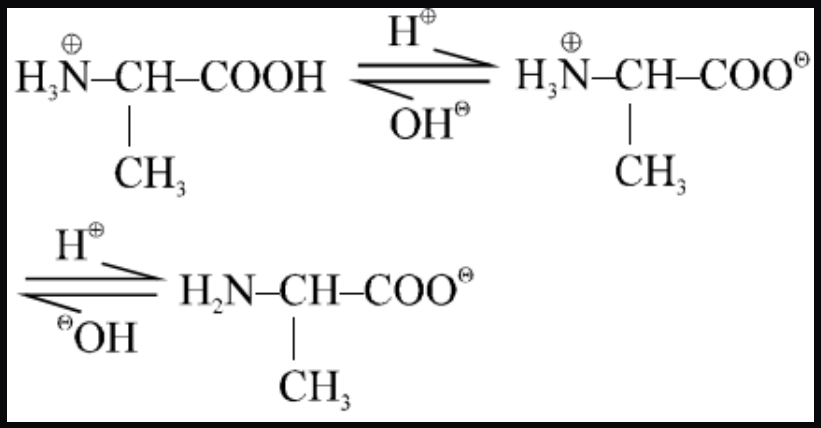
Q.33 Answer:

$D(+)-$ glucose and $D(-)-$ fructose give same osazone.

Solution:

Alanine zwitter ion is





Q.34 Answer:

I, II, III only

Solution:

(I, II) Properties of elements are periodic function of their atomic number. Number of non-metals are less as compared to number of metallic elements.

(III) In periodic table, first ionisation energy of elements along a period do not vary in regular manner due to

effective nuclear charge.

(IV) Ground state electronic configuration of $\mathrm{Pd}[Z=46]$ is $[\mathrm{Kr}]4d^{10}5s^{0}$.

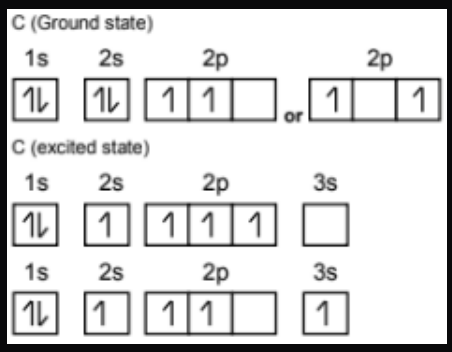
Hence, IVth statement is incorrect.

Thus, option (b) is correct.

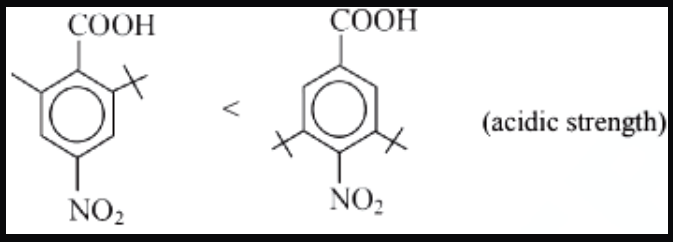
Q.35 Answer:

(i) & (iii)

Solution:

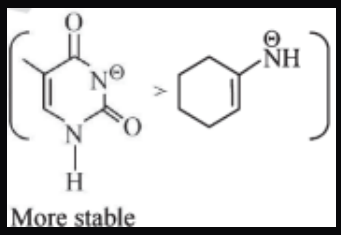


Q.36 Answer:



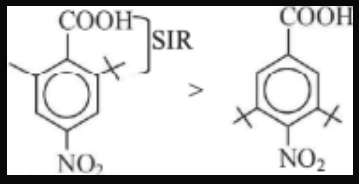
Solution:

(1) correct order $\rightarrow$



(2) correct order $2>1>3$

(3) correct order



(4) HOH order $\rightarrow$



C are incorrect

Q.37 Answer:

I, III only

Solution:

I. Key Idea $(-I)$ group, if are present along with $-COOH$ group in an acid, they increases the acidic property while (+) I groups or groups show hyperconjugation will decreases the acidic property.

(a) $NCCH_2COOH$: Contain, NC (i.e. $-I$) group along with $-COOH$, thus will increases the acidic nature.

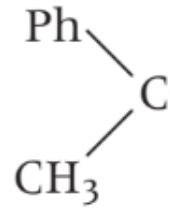
(b) FCH_2-COOH : Contain, F (i.e. $-I$) group thus increases the acidic nature. Between $NCCH_2COOH$ and $FCH_2COOH, NCCH_2COOH$ is more acidic due to stronger (-) I effect.

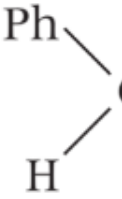
(c) H_3CCH_2COOH : Contain, CH_3 group which show hyperconjugation effect and this is least acidic. Hence, the given order is correct.

II) Key Idea (i) Structures which show more resonance structures or hyperconjugation among aldehydes are less reactive.

(ii) Aldehydes are more reactive than ketones.

a) $CH_3 \cdot CH_2 \cdot CHO$ is an aldehyde in which CH_3-CH_2- group show hyperconjugation effect.

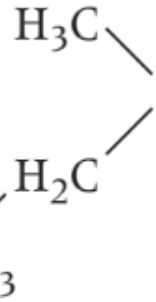
(b) $PhCOCH_3$, i.e.  CO is a ketone and is less reactive than $CH_3CH_2 \cdot CHO$ and PhCHO (as these are aldehydes).

(c) PhCHO, i.e.  CO is an aldehyde and is less

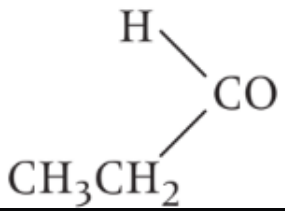
reactive than CH_3CH_2-CHO due to resonance effect of Ph-group.

Thus, correct order is $CH_3CH_2 \cdot CHO > PhCHO > PhCOCH_3$ and the given order is not correct.

(III) Key Idea Boiling point of alcohols (due to ability to form H-bond) are higher than that of aldehydes which are more than that of ketones. (due to less steric-hinderance and more surface area in aldehyde group, having same number of C-atoms)

(a) $H_3C \cdot COC_2H_5$,i.e.  CO is a ketone.

(b) $H_3C \cdot CH_2 \cdot CHO$ is an aldehyde, i.e.



(c) $H_3C - CH_2 - CH_2 - OH$ is an alcohol thus the given order for boiling point is correct. Hence, option (b) is the correct answer.

Q.38 Answer:

(a → iii), (b → iv), (c → ii), (d → v)

Solution:

$Co : 4s^2\ 3d^7$

$Cr : 4s^1\ 3d^5 \rightarrow \mathrm{Cr(CO)_6}$

$Cn : 4s^1\ 3d^{10} \rightarrow$ highest IE_2

$Zn : 4s^2\ 3d^{10} \rightarrow$ highest IE_3

$Ni : 4s^2\ 3d^8$

Q.39 Answer:

5 days

Solution:

Let wg of each be taken, then Initial mole of $\mathrm{P} = \frac{\mathrm{w}}{10}$; **mole of** $\mathrm{Q} = \frac{\mathrm{w}}{20}$

Final mole of $\mathrm{P} = \frac{\mathrm{w_1}}{5\times10}$;

Final mole of $Q = \frac{4w_1}{20\times5}$

$$\text{For P } \frac{\mathrm{N_{0p}}}{\mathrm{N_P}} = \mathrm{e}^{\lambda \mathrm{p}\times \mathrm{t}}$$

$$\text{For Q } \frac{\mathrm{N_{0_Q}}}{\mathrm{N_Q}} = \mathrm{e}^{\lambda \mathrm{Q}\times \mathrm{t}}$$

∴ For P $\frac{\mathrm{w}\times5\times10}{10\times\mathrm{w_1}} = \mathrm{e}^{\lambda \mathrm{p}\times 20}$ **...(i)**

For $Q\ \frac{W\times20\times5}{20\times w_1\times4} = \mathrm{e}^{\lambda_Q\times20}$ **...(ii)**

By Eqs. (i) and (ii) $4 = \mathrm{e}^{(\lambda_\mathrm{P}-\lambda_\mathrm{Q})\times20}$

$$\therefore 20\,(\lambda_P - \lambda_Q) = \log_e 4$$

$$\text{or } 20\left(\frac{0.693}{t_{1/2}} - \frac{0.693}{10}\right) = \log_e 4$$

$$\therefore t_{1/2} = 5$$

Q.40 Answer:

9

Solution:

$\mathrm{E^\circ_{cel}}$ à¤¸à¥ˆà¤² $= 0.76\mathrm{volt}$

Applying nernst equation

$$E_{\text{cel / à¤¸à¥‡à¤²}} = E^{o}_{\text{cel}} - \frac{0.0591}{2} \log \frac{\left[Zn^{2+}\right]\left[P_{H_2}\right]}{\left[H^{+}\right]^2}$$

$$0.26 = 0.76 - \frac{0.0591}{2} \log \frac{(0.1)(1)}{\left[H^{+}\right]^2}$$

$$\log \frac{0.1}{\left[H^{+}\right]^2} = \frac{2 \times 0.50}{0.0591}$$

or $\log 0.1 - \log \left[H^{+}\right]^2 = 17$

or $2pH = 17 - \log 0.1$

$pH = \frac{18}{2} = 9$

Q.41 Answer:

10 mole

Solution:

$$K_c = \frac{[H_2]\,[I_2]}{[HI]^2} = \frac{4 \times 4}{2^2} = 4$$

$2HI \rightleftharpoons H_2 + I_2$

$2 + 2x$

$(4 + a - x)$

$(4 - x) = 2 + 2x$

$x = 2/3$

Q.42 Answer:

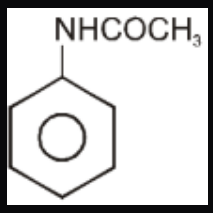
Mn^{3+} is oxidizing agent while Cr^{2+} is reducing agent

Solution:

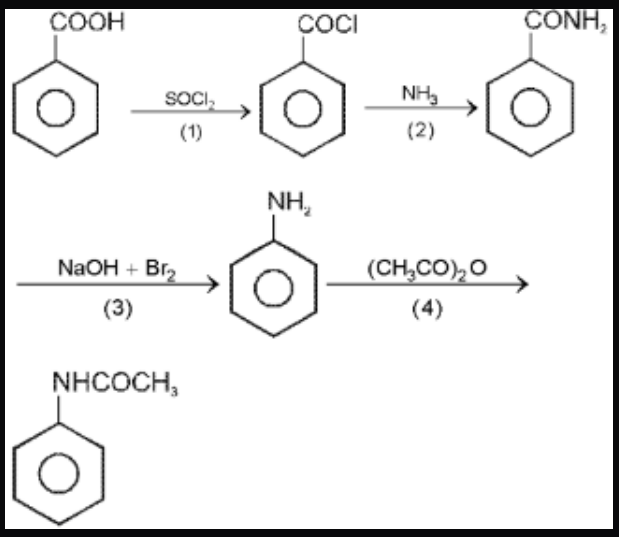
Mn is stable in +2 state so Mn^{3+} acts as an oxidizing agent by getting reduced to Mn^{2+} and attaining d^5 configuration.

Cr is stable in +3 state so Cr^{2+} acts as a reducing agent by getting oxidized to Cr^{3+}.

Q.43 Answer:



Solution:



Q.44 Answer:

0.26 V

Solution:

$\mathrm{Cr}\left|\mathrm{Cr}^{3+}(0.1\mathrm{M})\right\|\mathrm{Fe}^{2+}(0.01\mathrm{M})\left|\mathrm{Fe}\right.$

Oxidation half cell

$\mathrm{Cr} \longrightarrow \mathrm{Cr}^{3+} + 3\mathrm{e}^{-} \times 2$

Reduction half cell

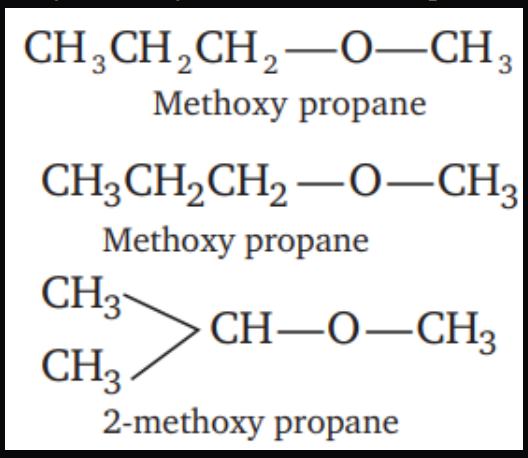
$\mathrm{Fe}^{2+} + 2e^{-} \longrightarrow \mathrm{Fe} \times 3$

Q.45 Answer:

methoxy propane

Solution:

Only two unsymmetrical ether is possible from the formula $\mathrm{C_4H_{10}O}viz.$



Q.46 Answer:

—

Solution:

$$\frac{(\Delta T)_P}{(\Delta T)_Q} = \frac{K_b \times \left(\frac{w_B \times 1000}{m_B \times w_A}\right)_P}{K_b \times \left(\frac{w_B \times 1000}{m_B \times w_A}\right)_Q}$$

$$\frac{0.2}{0.4} = \frac{\left(\frac{0.1 \times 1000}{100 \times 100}\right)_P}{\left(\frac{0.1 \times 1000}{100 \times m_B}\right)_Q}$$

$$\frac{1}{2} = \frac{(m_B)_Q}{100}$$

$$m_B = 50$$

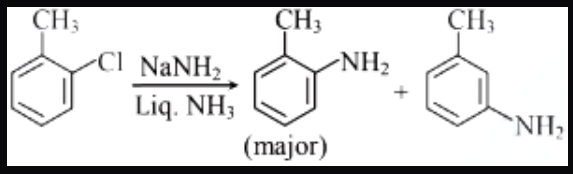
$$\therefore (m_B)_P : (m_B)_Q = 100 : 50$$

$$= 2 : 1$$

Q.47 Answer:

—

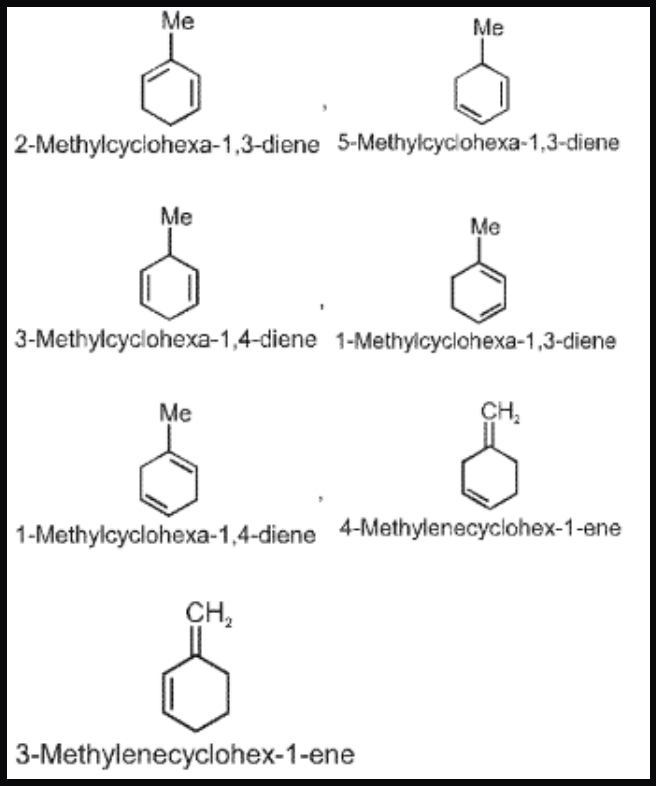
Solution:



Q.48 Answer:

—

Solution:

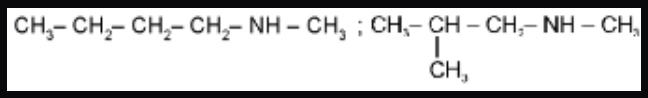


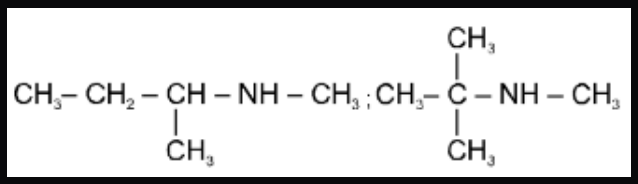
Q.49 Answer:

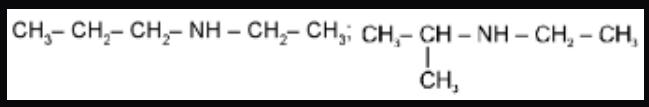
—

Solution:

$C_5H_{13}N$; D.U. $= 0$







Q.50 Answer:

—

Solution:

$$V = 2.18 \times 10^6 \times \frac{Z}{4} \text{ m/sec}$$

$$\Rightarrow 2.725 \times 10^6 = 2.18 \times 10^6 \times \frac{Z}{4}$$

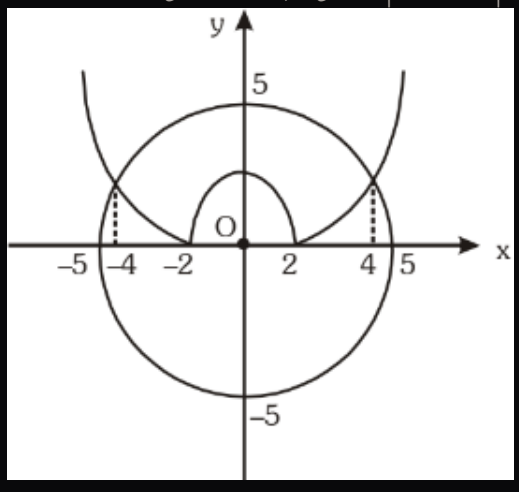
$$\Rightarrow Z = 5$$

Q.51 Answer:

$2 + \frac{25}{2}\sin^{-1}\left(\frac{4}{5}\right)$

Solution:

Here, $x^2 + y^2 = 25, 4y = |4 - x^2|$ could be sketched as, whose point of intersection could be obtained by



$$x^2+\frac{\left(4-x^2\right)}{16}=25$$
$$\Rightarrow\left(x^2-24\right)\left(x^2-16\right)=0$$
$$\Rightarrow x=\pm 4$$
$$\therefore \text{ Required area}$$
$$=\left\{\int_0^4\sqrt{25-x^2}dx-\int_0^2\left(\frac{4-x^2}{4}\right)dx-\int_2^4\left(\frac{x^2-4}{4}\right)dx\right.$$
$$=\left\{\left(\frac{x}{2}\sqrt{25-x^2}+\frac{25}{2}\sin^{-1}\left(\frac{x}{5}\right)\right)_0^4-\frac{1}{4}\left(4x-\frac{x^3}{3}\right)_0^2-\frac{1}{4}\left(\frac{x^3}{3}-4x\right)_2^4\right\}$$
$$=2+\frac{25}{2}\sin^{-1}\left(\frac{4}{5}\right)$$

Q.52 Answer:

$(-\infty, 0)$

Solution:

for f to be one-one $f'(x)>0$ and $f'(x)<0$ for all x
clearly f is continuous at $x=0$ and $f(0)=-1$
$x\leq 0, f'(x)=2(x+m)$ for $x<0$
$f'(x)$ can not be >0 for $\forall x<0$ if $m>0$
$\therefore \mathrm{f}'(\mathrm{x})<0$ for $\forall \mathrm{m}\leq 0$
but $\mathrm{m}\neq 0$ as for $\mathrm{x}>0$, f is constant and $\forall \mathrm{m}<0, \mathrm{f}'(\mathrm{x})<0, \forall \mathrm{x}>0$

Q.53 Answer:

$\frac{9}{100}$

Solution:

$$\mathrm{P}\left(\frac{\overline{\mathrm{A}}}{\mathrm{B}}\right)=\frac{2}{5}\Rightarrow\frac{\mathrm{P}(\overline{\mathrm{A}})\mathrm{P}(\mathrm{B})}{\mathrm{P}(\mathrm{B})}=\frac{2}{5}$$
$$\Rightarrow \mathrm{P}(\mathrm{A})=\frac{3}{5}$$
$$\mathrm{P}(\mathrm{A}\cap\mathrm{B})=\mathrm{P}(\mathrm{A})\mathrm{P}(\mathrm{B})=\frac{3}{5}\left(\frac{3}{4}-\frac{3}{5}\right)=\frac{9}{100}$$

Q.54 Answer:

351

Solution:

$\left(\frac{a^4+a^2+1}{a^2}\right)\left(\frac{b^4+7b^2+1}{b^2}\right)\left(\frac{c^4+11c^2+1}{c^2}\right)$

$\Rightarrow \left(a^2+\frac{1}{a^2}+1\right)\left(b^2+\frac{1}{b^2}+7\right)\left(c^2+\frac{1}{c^2}+11\right)$

$\left[\left(a-\frac{1}{a}\right)^2+3\right]\left[\left(b-\frac{1}{b}\right)^2+9\right]\left[\left(c-\frac{1}{c}\right)^2+13\right]$

clearly minimum value occurs when $\mathrm{a}^2=\mathrm{b}^2=\mathrm{c}^2=1$ and minimum value $=3\times9\times13=351$

Q.55 Answer:

-1

Solution:

Given that p_1, q_1, p_2q_2 are in A.P.

$\therefore (p_2-p_1)^2=(q_2-q_1)^2$

$\Rightarrow (p_2+p_1)^2-4p_1p_2=(q_2+q_1)^2-4q_1q_2$

$\Rightarrow \left(\frac{-b}{a}\right)^2-4\left(\frac{c}{a}\right)=\left(\frac{-b}{c}\right)^2-4\left(\frac{a}{c}\right)$

$\Rightarrow \frac{b^2-4ac}{a^2}=\frac{b^2-4ac}{c^2}$

Since b^2-4ac is the discriminant of both the equations and roots are different

$\therefore \quad \mathrm{b}^2\neq4\mathrm{ac}$

$\therefore \quad \mathrm{a}^2=\mathrm{c}^2$

$\Rightarrow \mathrm{a}=\mathrm{c}$ (Not possible because two quadratic equations become identical)

or $\mathrm{a}=-\mathrm{c}$

$\Rightarrow \frac{\mathrm{a}}{\mathrm{c}}=-1$

Q.56 Answer:

18

Solution:

$\mathrm{Lim}_{x\to1}\frac{\sin^2\left(x^3+x^2+x-3\right)}{\left(x^3+x^2+x-3\right)^2}\cdot\frac{\left(x^3+x^2+x-3\right)^2}{1-\cos\left(x^2-4x+3\right)}$

$=\ (1)\ \mathrm{Lim}_{x\to1}\frac{\left(x^2-4x+3\right)^2}{1-\cos\left(x^2-4x+3\right)}\cdot\frac{\left(x^3+x^2+x-3\right)^2}{\left(x^2-4x+3\right)^2}$

$=$ (1) (2) $\mathrm{Lim}_{x\to1}\left(\frac{x^3+x^2+x-3}{x^2-4x+3}\right)^2=2l^2$ where $I=\mathrm{Lim}_{x\to1}\frac{3x^2+2x+1}{2x-4}$ (using L'Hospital's rule)

$=\frac{6}{-2}=-3$

$\therefore \quad \mathrm{L}=2(-3)^2=18$ Ans.

Q.57 Answer:

2

Solution:

$\mathrm{A}^2 = \mathrm{I}$

$\Rightarrow A^4 = A^6 = \ldots\ldots\ldots = I$

Now $A^x = I$

$\Rightarrow x = 2, 4, 6, 8, \ldots\ldots\ldots$

$\sum_x (\cos^x \theta + \sin^x \theta) = \cos^2 \theta + \cos^4 \theta + \ldots\ldots\ldots + \sin^2 \theta + \sin^4 \theta + \sin^6 \theta + \ldots\ldots\ldots$

$= \frac{\cos^2 \theta}{1-\cos^2 \theta} + \frac{\sin^2 \theta}{1-\sin^2 \theta} = \cot^2 \theta + \tan^2 \theta$

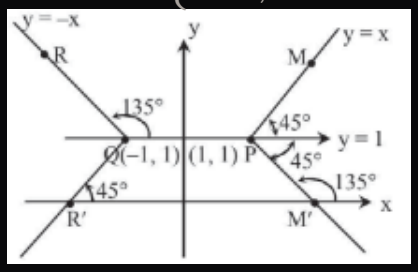
Which has minimum value 2.

Q.58 Answer:

$y = -|x| + 2$

Solution:

$$y = |x| = \begin{cases} x; x > 0 \\ -x; x < 0 \end{cases}$$



Lines PM **and** QR **are a pair of rays.**

Lines PM' **and** QR' **are pair of image lines w.r.t. line** $\mathrm{y} = 1$.

$\Rightarrow$ **Equation of line** PM'

$y - 1 = (-1)(x - 1) \Rightarrow y = -x + 2$ **...(1)**

$\Rightarrow$ **Equation of line** QR'

$y - 1 = (1)(x + 1) \Rightarrow y = x + 2$ **...(2)**

Combined equation of (1) and (2)

$\Rightarrow y = |x| + 2$

Q.59 Answer:

64

Solution:

$$\mathrm{A}^{-1} + \mathrm{B}^{-1} = 2\mathrm{I}$$

$$\Rightarrow \mathrm{I} + \mathrm{AB}^{-1} = 2\ \mathrm{A}$$

$$\Rightarrow \mathrm{B} + \mathrm{A} = 2\mathrm{AB} \text{ or } 2\mathrm{AB} = \mathrm{I}$$

Q.60 Answer:

$x(x - y) = y \log (\mathrm{Ce}^x - 1)$

Solution:

$$x^2\frac{dy}{dx} - 2y(x-y) + y^2e^xe^{-x^2/y} = 0$$

$$\frac{x^2}{y^2}\frac{dy}{dx} - \frac{2(x-y)}{y} = -e^xe^{-x^2/y}$$

$$e^{x^2/y}\frac{x^2}{y^2}\frac{dy}{dx} - \frac{2x}{y}e^{x^2/y} + 2e^{x^2/y} + e^x = 0$$

Let $\mathrm{e}^{\mathrm{x}^2/\mathrm{y}} = \mathrm{t}$

$$\mathrm{e}^{\mathrm{x}^2/\mathrm{y}}\left(\frac{2\mathrm{xy} - \mathrm{x}^2\frac{\mathrm{dy}}{\mathrm{dx}}}{\mathrm{y}^2}\right) = \frac{\mathrm{dt}}{\mathrm{dx}}$$

$$\Rightarrow \mathrm{e}^{\mathrm{x}^2/\mathrm{y}}\frac{2\mathrm{x}}{\mathrm{y}} - \frac{\mathrm{x}^2}{\mathrm{y}^2}\mathrm{e}^{\mathrm{x}^2/\mathrm{y}}\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{\mathrm{dt}}{\mathrm{dx}}$$

$$-\frac{\mathrm{dt}}{\mathrm{dx}} + 2\mathrm{t} + \mathrm{e}^{\mathrm{x}} = 0 \Rightarrow \frac{\mathrm{dt}}{\mathrm{dx}} - 2\mathrm{t} = \mathrm{e}^{\mathrm{x}}$$

On solving, $t = -e^x + Ce^{2x}$ **where** C **is constant of integration**

$$\text{or } \mathrm{e}^{\frac{\mathrm{x}^2}{\mathrm{y}}-\mathrm{x}} = -1 + C\mathrm{Ce}^{\mathrm{x}}$$

$$\left(\frac{\mathrm{x}^2}{\mathrm{y}} - \mathrm{x}\right) = \log\left(\mathrm{Ce}^{\mathrm{x}} - 1\right)$$

$$\mathrm{x}(\mathrm{x}-\mathrm{y}) = \mathrm{y}\log\left(\mathrm{Ce}^{\mathrm{x}} - 1\right)$$

Q.61 Answer:

$\left(2, \frac{9}{4}\right)$

Solution:

Both roots of the equation $x^2 - x + k - 2 = 0$ **must be positive and distinct.**

$$\therefore \mathrm{D} > 0 \Rightarrow 1 - 4(\mathrm{k}-2) > 0 \Rightarrow \mathrm{k} < \frac{9}{4}$$

$$\alpha\beta > 0 \Rightarrow \mathrm{k} - 2 > 0 \Rightarrow \mathrm{k} > 2$$

$$\alpha + \beta > 0 \Rightarrow \text{ true}$$

$$\therefore \mathrm{k} \in \left(2, \frac{9}{4}\right)$$

Q.62 Answer:

5

Solution:

$\text{Put } y=\dfrac{1-\alpha}{1+\alpha} \Rightarrow \alpha=-\left(\dfrac{y-1}{y+1}\right)$

$\therefore -\left(\dfrac{y-1}{y+1}\right)^3-\left(\dfrac{y-1}{y+1}\right)+3=0$

$\Rightarrow -(y-1)^3-(y-1)(y+1)^2+3(y+1)^3=0 \text{ have}$

$\text{root as } \dfrac{1-\alpha}{1+\alpha}, \dfrac{1-\beta}{1+\beta}, \dfrac{1-\gamma}{1+\gamma}$

$\Rightarrow -\left(y^3-3y^2+3y-1\right)-\left(y^2-1\right)(y+1)$

$\quad +3\left(y^3+3y^2+3y+1\right)=0$

$\Rightarrow -\left(y^3-3y^2+3y-1\right)-\left(y^3+y^2-y-1\right)$

$\Rightarrow y^3+11y^2+7y+5=0$

$|\text{product}|=|-5|=5$

Q.63 Answer:

a_{51}

Solution:

Mean square deviation is minimum about mean i.e. a_{51} **and median of the group is** a_{51}

$\Rightarrow$ **Mode will also be** a_{51}

Q.64 Answer:

$-\mathrm{e}^{-1}$

Solution:

Given $\int_0^{\mathrm{x}} f(\mathrm{t})\mathrm{dt}+\int_0^{\mathrm{x}}(\mathrm{x}-\mathrm{t})\cdot f(\mathrm{t})\mathrm{dt}=\mathrm{e}^{-\mathrm{x}}-1$ **or,** $\int_0^{\mathrm{x}} f(\mathrm{t})\mathrm{dt}+\mathrm{x}\int_0^{\mathrm{x}} f(\mathrm{t})\mathrm{dt}-\int_0^{\mathrm{x}} \mathrm{t}f(\mathrm{t})\mathrm{dt}=\mathrm{e}^{-\mathrm{x}}-1$ **Diff. both sides with respect to** x**, we get.**

$f(\mathrm{x})+\mathrm{x}.f(\mathrm{x})+\int_0^{\mathrm{x}} f(\mathrm{t})\mathrm{dt}-\mathrm{x}f(\mathrm{x})=-\mathrm{e}^{-\mathrm{x}}$

$\Rightarrow f(\mathrm{x})+\int_0^{\mathrm{x}} f(\mathrm{t})\mathrm{dt}=-\mathrm{e}^{-\mathrm{x}}$

again diff. w.r.to $\mathrm{x} \Rightarrow \mathrm{e}^{\mathrm{x}}\left(f'(\mathrm{x})+f(\mathrm{x})\right)=1$

$\mathrm{e}^{\mathrm{x}}\cdot f(\mathrm{x})=\mathrm{x}+\mathrm{c}$, **from** $(1), f(0)=-1.$

$\therefore f(\mathrm{x})=(\mathrm{x}-1)\mathrm{e}^{-\mathrm{x}}$

$\therefore \int_0^1 f(\mathrm{x})\mathrm{dx}=\int_0^1(\mathrm{x}-1)\mathrm{e}^{-\mathrm{x}}\mathrm{dx}=\left(-(\mathrm{x}-1)\mathrm{e}^{-\mathrm{x}}\right)_0^1+\int_0^1 \mathrm{e}^{-\mathrm{x}}\mathrm{dx}$

$=-1-\left(\mathrm{e}^{-\mathrm{x}}\right)_0^1=-1-\left(\frac{1}{\mathrm{e}}-1\right)=-\frac{1}{\mathrm{e}}$

Q.65 Answer:

$\frac{3}{\sqrt{35}}$

Solution:

$\overrightarrow{\mathrm{r}}=(-2,-1,-2)+\lambda(1,-2,1)$

$\overrightarrow{\mathrm{r}}=(-1,0,-1)+\mathrm{k}(2,1,1)$

$\text{S.D } =\left|\dfrac{(\hat{i}+\hat{j}+\hat{k})\cdot(\hat{i}-2\hat{j}+\hat{k})\times(2\hat{i}+\hat{j}+\hat{k})}{|(\hat{i}-2\hat{j}+\hat{k})\times(2\hat{i}+\hat{j}+\hat{k})|}\right|=\dfrac{3}{\sqrt{35}}$

Q.66 Answer:

26

Solution:

$$\cos^{-1}(2x)+\cos^{-1}(3x)=\pi-\cos^{-1}(x)=\cos^{-1}(-x)$$
$$\cos^{-1}\left[(2x)(3x)-\sqrt{1-4x^2}\sqrt{1-9x^2}\right]=\cos^{-1}(-x)$$
$$6x^2-\sqrt{1-4x^2}\cdot\sqrt{1-9x^2}=-x$$
$$\left(6x^2+x\right)^2=\left(1-4x^2\right)\left(1-9x^2\right)$$
$$\Rightarrow x^2+12x^3=1-13x^2$$
$$\Rightarrow 12x^3+14x^2-1=0$$
$$\therefore a=12; b=14; c=0$$
$$\Rightarrow a+b+c=26 \text{ Ans.}$$

Q.67 Answer:

$2\sqrt{2}$

Solution:

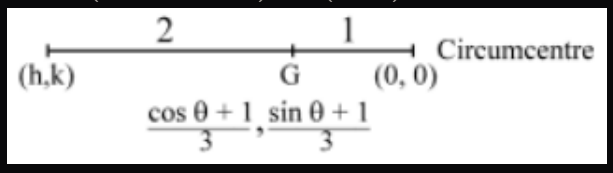
$$|\vec{a}||\vec{b}|\sin\theta+|\vec{a}||\vec{b}|\cos\theta=\sqrt{2}|\vec{a}||\vec{b}|$$
$$\Rightarrow \sin\theta+\cos\theta=\sqrt{2}$$
$$\Rightarrow \quad \theta=\frac{\pi}{4}$$
$$\text{Area of ABCD}=|\vec{a}\times\vec{b}|=2\sqrt{2}$$

Q.68 Answer:

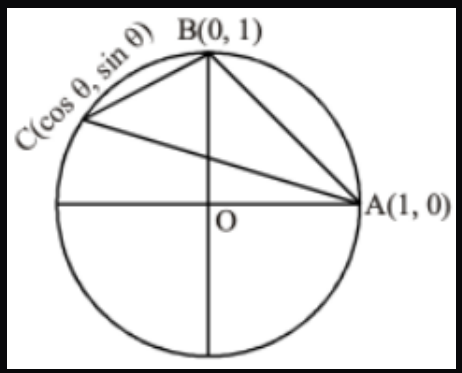
$x^2+y^2-2x-2y+1=0$

Solution:

Let $C(\cos\theta,\sin\theta); H(h,k)$ is the orthocentre of the $\triangle ABC$



$$h=1+\cos\theta$$
$$k=1+\sin\theta$$



$$(x-1)^2+(y-1)^2=1$$
$$x^2+y^2-2x-2y+1=0$$

Q.69 Answer:

$\frac{1}{84}$

Solution:

$f(\mathrm{x}) = |\mathrm{x}-1| + 2\left|\mathrm{x}-\frac{1}{2}\right| + \ldots\ldots + 119\left|\mathrm{x}-\frac{1}{119}\right|$

minimum value occurs at median

Total number of mod

$= 1 + 2 + \ldots + 119 = 7140$

$\dfrac{\mathrm{n}(\mathrm{n}+1)}{2} = 3570$

$\mathrm{n} = 89$

minimum occurs at $x = \frac{1}{84}$

Q.70 Answer:

$y = \ln\left(\tan\left(y - e^{\tan^{-1}x} + C\right)\right)$

Solution:

$$\frac{e^{\tan^{-1}x}}{(1+x^2)} \cdot dx = \frac{\left[e^y + (e^y-1)^2\right] \cdot dy}{1+e^{2y}}$$

$$\int \frac{e^{\tan^{-1}x}}{1+x^2} \cdot dx = \int \frac{e^{2y} - e^y + 1}{1+e^{2y}} \cdot dy = \int 1 - \frac{e^y}{1+(e^y)^2} \cdot dy$$

$$e^{\tan^{-1}x} = y - \tan^{-1}e^y + c.$$

$$\tan^{-1}e^y = y - e^{\tan^{-1}x} + c$$

$$y = \ln\left[\tan\left(y - e^{\tan^{-1}x} + c\right)\right]$$

Q.71 Answer:

—

Solution:

At $x \to 0^+, \{\} = .$

$$\lim_{x\to 0^+} \frac{x^x \cos\left(x^{\sin x+1}\right)}{\cos(\tan x - x)\cos(x-x)} = k$$

$k = 1$, as $\lim\limits_{x\to 0^+} x^x = 1$

Q.72 Answer:

—

Solution:

$\underline{2015} + 3^{2015}$

2015 has last two digits zero.

$3^{2015} \equiv 3.\left(3^{2014}\right)$

$\equiv 3.9^{1007} = (3)(10-1)^{1007}$

on expansion last two digits $\equiv 07$

Q.73 Answer:

—

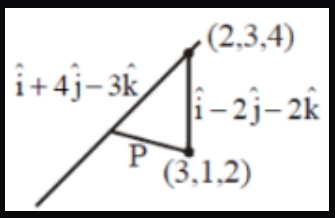
Solution:

L_1 and L_2 are intersecting

$\Rightarrow k = 3$ and their point of intersection is $(2, 3, 4)$.

$$L_3 : \frac{x-2}{1} = \frac{y-3}{4} = \frac{z-4}{-3}$$

$$P = \frac{1(\hat{i} - 2\hat{j} - 2\hat{k}) \times (\hat{i} + 4\hat{j} - 3\hat{k})}{|\hat{i} + 4\hat{j} - 3\hat{k}|} = \frac{\sqrt{233}}{\sqrt{26}}$$



$$[P] = [\sqrt{8 \cdot 9}] = 2$$

Q.74 Answer:

—

Solution:

$$\frac{z + \overline{z}}{2} = |z - 1|$$

$$\Rightarrow x^2 = (x-1)^2 + y^2 \text{ or } y^2 = 2x - 1$$

$$\text{Let } z_1 \left(\frac{t_1^2 + 1}{2}, t_1\right) \& z_2 \left(\frac{t_2^2 + 1}{2}, t_2\right)$$

$$\because \arg(z_1 - z_2) = \frac{\pi}{3} \Rightarrow \frac{t_2 - t_1}{\frac{t_2^2 - t_1^2}{2}} = \sqrt{3}$$

$$\Rightarrow \frac{2}{t_1 + t_2} = \sqrt{3}$$

$$\Rightarrow \text{Im}(z_1 + z_2) = \frac{2}{\sqrt{3}}$$

Q.75 Answer:

—

Solution:

$$2I_1 = \int_0^{\pi} \frac{2 \sin 884x \sin 1122x}{\sin x} dx$$

$$= \int_0^{\pi} \frac{\cos 238x}{\sin x} dx - \int_0^{\pi} \frac{\cos 2006x}{\sin x} dx$$

Let $I_{2m} = \int_0^{\pi} \frac{\cos 2mx}{\sin x} dx$ then

$$I_{2m} - I_{2m-2} = \int_0^{\pi} \frac{\cos 2mx - \cos(2m-2)x}{\sin x} dx$$

$$= 2\left(\frac{\cos(2m-1)x}{2m-1}\right)_0^{\pi} = -\frac{4}{2m-1}$$

$$\therefore \quad I_{2006} - I_2 = -4\left(\frac{1}{3} + \frac{1}{5} + \ldots + \frac{1}{2005}\right)$$

$$\& I_{238} - I_2 = -4\left(\frac{1}{3} + \frac{1}{5} + \ldots + \frac{1}{237}\right)$$

$$2I_1 = 4\left(\frac{1}{239} + \frac{1}{241} + \ldots + \frac{1}{2005}\right)$$

Now

$$\mathrm{I_2} = \int_0^1 \left(\mathrm{x}^{238} + \mathrm{x}^{240} + \ldots + \mathrm{x}^{2004}\right) \mathrm{dx} = \left(\frac{1}{239} + \frac{1}{241} + \ldots\right)$$

$$\therefore \quad \frac{\mathrm{I_1}}{\mathrm{I_2}} = 2$$